

Phenix Developer Reference

1 documentation pages

Generated from local Phenix documentation

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1. AI-Powered Phenix & CCTBX Programming

AI-Powered Phenix & CCTBX Programming

developer.html

Why use AI: The Phenix developer AI chatbot has all the Phenix and CCTBX documentation and code in its sources. It can help you get going quickly. It can even write unit tests for you.

Be aware: AI chatbots usually do not get everything right on the first try. It is your job to tell them what is wrong (for starters, paste in a traceback!) and to make your code work just the way you want it to.

Be safe: Look up all packages before you install them with pip to make sure they are real. Read the code from AI scripts before running them. Make sure they make sense and do not do anything bad.

Use the right AI tool for the job: Use the Phenix developer chat to write Phenix and CCTBX code. It knows the code base and how to use the Program Template. Use Gemini or other general-purpose AI chatbots to write scripts that do not use Phenix or CCTBX tools or to write HTML. These chatbots are better at programming but they do not know about the Phenix code base.

Chatbots

Phenix developer chat **Private, do not share link**

Use this for Phenix or CCTBX programming questions or to write Python scripts in Phenix/CCTBX. Designed to allow easy creation of ProgramTemplate files.

CCTBX developer chat

Use this for CCTBX programming questions or scripts.

Phenix user help chat

Use this for questions on how to run Phenix or for general questions about crystallography and cryo-EM structure determination.

Hints on using AI programming chatbots, adding docstrings with AI, and on using the Phenix user chatbot

Programming Documentation

CCTBX developer documentation

Comprehensive guide to using CCTBX tools. High-level cctbx managers, maps, and low-level cctbx objects and tools. Includes tutorials, sample scripts, how to test your code, and many helpful hints on using CCTBX tools.

Phenix developer documentation

How to use the Program Template, how to use the model-building tools in Phenix, handy restraint jiffies, Phenix/CCTBX high-level reference guide.

PHENIX/CCTBX Programming API

Fully indexed documentation of all CCTBX and Phenix Python classes and methods. Includes all code from CCTBX and from Phenix. You can update this on your machine with the commands:

```
phenix.rebuild_phenix_api
```

CCTBX Programming API

Fully indexed documentation and code for all CCTBX Python classes and methods. You can update this on your machine with the commands:

```
phenix.rebuild_cctbx_api
```